CSE 4345: Big Data Analytics

Week 6, Part 2 — Research Article & Case Study

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Part 2 Overview — Week 6

Today's Agenda

1. Paper 1 — Zaharia et al. (2010): Spark HotCloud
2. Paper 2 — Zaharia et al. (2012): RDDs at NSDI
3. Case Study — Databricks Daytona GraySort (2013)
4. Live Exercise

CLO Addressed

CLO3 — Explain in-memory distributed processing
PLO(b,c)

Research Articles Covered

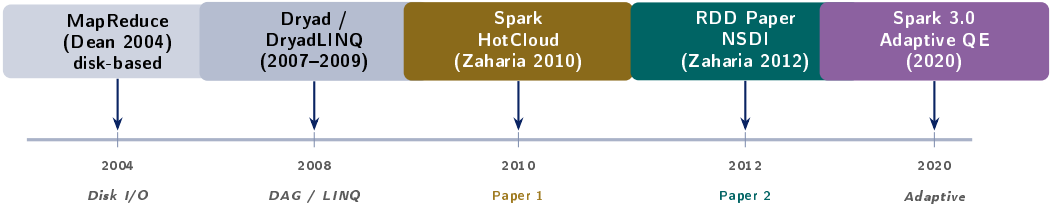
1. Zaharia, M., et al. (2010). *Spark: Cluster Computing with Working Sets.* USENIX HotCloud 2010.
2. Zaharia, M., et al. (2012). *Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing.* USENIX NSDI 2012.

Case Study

Databricks Daytona GraySort (2013): 100 TB sorted in **23 minutes** on 206 EC2 nodes — 3× faster than Hadoop on 3.5× fewer machines.

Paper 1 — Zaharia et al. (2010): Spark HotCloud

Research Landscape: In-Memory Cluster Computing



The Fundamental Problem with MapReduce

Machine learning and graph algorithms are **iterative**: each iteration reads the output of the previous one. MapReduce writes every intermediate result to HDFS. For k iterations: $k\times$ HDFS read + $k\times$ HDFS write. On spinning disk at 100 MB/s, 10 iterations over 100 GB costs $\approx$**3 hours of I/O alone**. Spark loads the dataset into RAM *once* and iterates in memory.

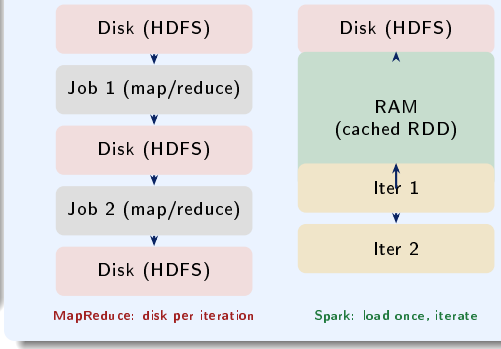
Zaharia et al. (2010): HotCloud Publication Context

Full Citation

Zaharia, M., Chowdhury, M., Franklin, M. J., Shenker, S., & Stoica, I. (2010). **Spark: Cluster Computing with Working Sets.** In *Proc. 2nd USENIX Workshop on Hot Topics in Cloud Computing (HotCloud '10)*. Boston, MA, USA.

Venue: USENIX HotCloud (workshop, short paper)
Cited by: >5,000 (Google Scholar, 2024)
Authors: Matei Zaharia (PhD student, UC Berkeley AMP Lab), co-advised by Scott Shenker & Ion Stoica

Working Set vs. Streaming



The Key Observation

Many real workloads operate on a **working set** — a dataset that is *reused across multiple parallel operations*:

Paper 2 — Zaharia et al. (2012): RDDs at NSDI

Zaharia et al. (2012): NSDI Publication Context

Full Citation

Zaharia, M., Chowdhury, M., Das, T., Dave, A., Ma, J., McCauley, M., Franklin, M. J., Shenker, S., & Stoica, I. (2012). **Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing.** In *Proc. 9th USENIX Symposium on Networked Systems Design and Implementation (NSDI '12)*, pp. 15–28. San Jose, CA, USA.

Venue: USENIX NSDI — Rank A* (CORE), top systems venue
Cited by: >13,000 (Google Scholar, 2024)
Best Paper Award: NSDI 2012

From Workshop to Full Paper

The 2010 HotCloud paper was a **4-page proof-of-concept**. The 2012 NSDI paper introduced **RDDs** as a formal abstraction, provided a full

The RDD Definition

*"A **Resilient Distributed Dataset** is a read-only, partitioned collection of records that can be created through deterministic operations on data in stable storage or other RDDs."*

- **Resilient**: lost partitions are recomputed from lineage, not replicated checkpoints
- **Distributed**: partitions spread across nodes
- **Dataset**: a collection of records, not a mutable key-value store

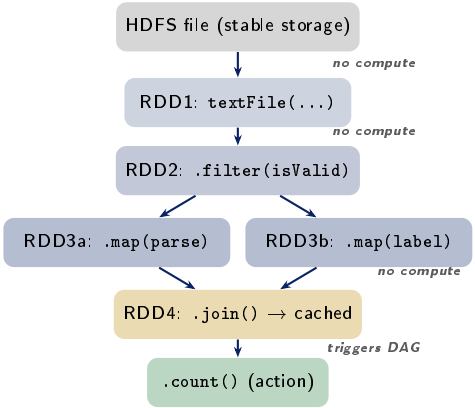
Technical Contribution 1 — RDD Lineage & Fault Tolerance

Lineage Graph: The Key Innovation

RDDs do **not replicate data** to tolerate failures. Instead, each RDD records a **lineage**: a log of the transformations that produced it. If a partition is lost, Spark re-executes only those transformations on the relevant input partitions.

Two types of operations:

- **Transformations** (lazy): `map`, `filter`, `flatMap`, `groupByKey`, `reduceByKey`, `join` — return a new RDD, do *not* trigger computation
- **Actions** (eager): `count`, `collect`, `reduce`, `saveAsTextFile` — trigger a DAG execution from root to output



Technical Contribution 2 — Narrow vs. Wide Dependencies

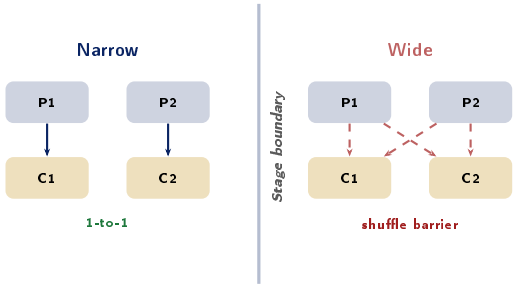
Dependency Classification

Narrow dependency: each parent partition contributes to *at most one* child partition.

- Examples: `map`, `filter`, `union`
- No data shuffle required
- Lost partition recomputed from **one** parent partition
- Pipelined within a single **Stage** (no barrier)

Wide dependency: each parent partition may contribute to *multiple* child partitions.

- Examples: `groupByKey`, `reduceByKey`, `sortByKey`, `join` (on un-partitioned RDDs)
- Requires a **shuffle** — network-intensive barrier
- Marks a Stage boundary in the DAG



Wide dep. ⇒ shuffle ⇒ new Stage

Technical Contribution 3 — Persistence & Partitioning

Persistence Levels (from the paper)

StorageLevel	Meaning
`MEMORY_ONLY`	JVM heap; spill re-computes
`MEMORY_AND_DISK`	Heap; overflow to local disk
`DISK_ONLY`	Local disk; slower reuse
`MEMORY_ONLY_SER`	Kryo-serialised bytes in heap; smaller but slower
`OFF_HEAP`	Native memory (Unsafe); avoids GC pressure

When to Cache?

Cache an RDD if it is **reused in >1 action** (iterative ML, interactive queries). Caching single-use RDDs wastes RAM and triggers eviction of other cached data (LRU eviction policy).

Partitioning for Key-Value RDDs

- **Hash partitioner** (default): `key.hashCode() % numPartitions`
  Uniform for random keys; can hot-spot on skewed data
- **Range partitioner**: samples key distribution, assigns ranges to partitions; used by `sortByKey()`
- **Custom partitioner**: user-supplied; co-partition two RDDs on the same key to eliminate shuffle on join

Impact: Benchmark Results (RDD paper)

Logistic regression (10 iterations):

- Hadoop MapReduce: 127 s/iteration (disk)
- Spark (MEMORY_ONLY): **0.96 s/iteration**
- Speedup: 132×

K-means clustering (10 iterations): Hadoop 150 s vs. Spark 5 s = 30×

Impact Today — Zaharia et al. (2010, 2012)

What Spark Built on the RDD Foundation

- **Spark SQL / DataFrames** (2014): RDDs wrapped in a typed columnar schema; **Catalyst optimizer** rewrites logical plans; **Tungsten** generates JVM bytecode for execution
- **Spark Streaming / Structured Streaming** (2013/2016): micro-batch or continuous processing via DStreams (RDDs) and then DataStreamReader (DataFrames)
- **MLlib / ML Pipelines** (2014): iterative ML on cached DataFrames; Pipeline API mirrors scikit-learn's `fit/transform`
- **GraphX** (2014): RDD-backed property graphs for PageRank, label propagation, connected components
- **Delta Lake / Iceberg integration** (2019+): Spark is the primary compute engine for ACID lakehouse architectures

CLO3 Connection

RDDs + Catalyst + Tungsten directly address CLO3: *how* Spark explains and optimises distributed in-memory computation.

Case Study — Databricks Daytona GraySort (2013)

The GraySort Benchmark

What is Daytona GraySort?

GraySort (named after Jim Gray, Turing Award 1998) is the **world-record sorting competition** for large-scale data systems. Two categories:

- **Daytona**: real-world rules — commodity hardware, standard sort record format (100-byte records, 10-byte key), fault tolerance required, no code specialisation; the prestigious category
- **Indy**: custom hardware / software optimisations allowed

Benchmark dataset: 100 TB of random 100-byte records (10^{12} records).

Metric: wall-clock time to sort, verify, and write output.

Previous Record (Yahoo!, 2009)

Yahoo! sorted 100 TB on a **3,800-node Hadoop MapReduce** cluster in **173 minutes**. In 2012, a team at Yahoo! improved the Hadoop record to **72 minutes** using 2,100 nodes.

Key bottleneck: MapReduce's sort phase writes intermediate data to disk between the map and reduce stages — twice for each byte (spill + merge). At 100 TB, this is ≈200 TB of additional disk I/O.

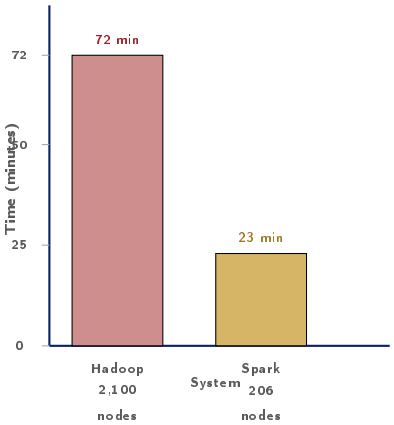
Why Sort?

Sorting exercises every component of a distributed system: network bandwidth, disk I/O, CPU (comparison), memory, and distributed coordination. A faster sort implies a faster data platform in general.

Databricks Result: 100 TB in 23 Minutes (November 2013)

The Databricks Setup

- **Cluster**: 206 EC2 i2.8xlarge instances on Amazon Web Services
- **Per node**: 32 vCPUs, 244 GB RAM, 8×800 GB SSD (local ephemeral storage)
- **Total RAM**: $206 \times 244 = 50.3$ TB
- **Total SSD**: $206 \times 6.4 = 1.3$ PB
- **Network**: 10 Gbps per node
- **Input**: 100 TB of random 100-byte records stored in HDFS (3× replication = 300 TB on disk)
- **Spark version**: 0.8 (pre-DataFrame era)



Result

100 TB sorted in 23 minutes

4.27 TB/min throughput

Key Ratios

- 3.1× faster than Hadoop (72 min → 23 min)

Why Spark Won: Technical Analysis

What Made Spark Faster at GraySort?

1. **Sort-based shuffle** (not hash-based): Spark's sort shuffle writes sorted runs to disk once, then merges — MapReduce's hash shuffle spills, merges, and re-reads multiple times. At 100 TB, this eliminates ≈120 TB of extra I/O.
2. **SSD-aware I/O**: Databricks configured Spark to use all 8 SSDs as shuffle directories simultaneously — 8× the sequential write bandwidth of one SSD.
3. **Off-heap sort buffers**: Spark's `UnsafeShuffleWriter` uses sun.misc.Unsafe to sort records in native memory, bypassing JVM GC pauses during the sort phase.
4. **Fewer stages**: the sort is a single wide transformation (`sortByKey`) — no intermediate HDFS writes between map and reduce.

Databricks Production Spark ML (2014)

Beyond sorting, Databricks published results showing Spark MLlib outperforming Mahout (MapReduce-based ML):

- **Logistic regression** on 1 TB: Spark 2 min vs. Mahout 10+ hours
- **Collaborative filtering** (ALS) on Netflix data: Spark 35 min vs. MR implementation 10 hours
- The speedup comes *primarily* from keeping the model and working set in RAM across iterations

Long-Term Impact

The GraySort result generated enormous press coverage and directly drove Spark's adoption: Apache Spark went from 200 GitHub stars in 2013 to >36,000 by 2020; Databricks raised $150M Series B within 6 months of the record.

Lessons Learned & Discussion Questions

Key Lessons from Spark's Rise

1. **In-memory is a multiplier, not a panacea**: Spark is 10–100× faster on iterative workloads; for single-pass ETL over data larger than RAM, gains are smaller (2–5×)
2. **The lineage model scales elegantly**: re-computing a lost partition from lineage is cheaper than maintaining 3× replicated in-memory state
3. **Shuffle is the bottleneck**: both at GraySort and in production; minimising wide dependencies (shuffle) is the primary Spark tuning skill
4. **Benchmark marketing is real**: GraySort made Spark famous; it also hid cases where Spark's JVM memory overhead hurt performance (many small files, high-cardinality `groupBy`)
5. **Open-source velocity**: the Apache Spark community grew to 1,000+ contributors in 5 years; Databricks' business model (cloud + support) aligned commercial incentives with open-source development

Live Exercise

Live Exercise — Spark RDD Lineage in PySpark

Task (25 minutes, pairs)

Run the following PySpark script locally (Spark local mode).

Goal: build and inspect an RDD lineage; observe lazy evaluation; compare cached vs. uncached timing.

Questions:

1. Call `rdd.toDebugString()` before and after `cache()` — what changes?
2. Time `rdd.count()` twice without `cache()`. Then add `.cache()` and time again. Report the speedup.
3. Change `reduceByKey` to `groupByKey`. Use the Spark UI (localhost:4040) to compare shuffle read/write bytes between the two.

PySpark Skeleton

```
from pyspark import SparkContext
import time
sc = SparkContext("local[4]", "RDDDemo")
sc.setLogLevel("WARN")
# Build lineage (lazy - no compute yet)
lines = sc.textFile("README.md")
words = lines.flatMap(lambda l: l.split())
pairs = words.map(lambda w: (w.lower(), 1))
counts = pairs.reduceByKey(lambda a, b: a+b)
# Inspect lineage BEFORE action
print(counts.toDebugString().decode())
# First action - triggers full DAG
t0 = time.time()
n  = counts.count()
print(f"count={n}  time={time.time()-t0:.2f}s")
# Cache and re-run
counts.cache()
counts.count()          # warm up cache
t1 = time.time()
n  = counts.count()     # served from cache
print(f"cached count={n}  "
      f"time={time.time()-t1:.4f}s")
sc.stop()
```

References

Research Articles

- Zaharia, M., et al. (2010). Spark: Cluster computing with working sets. *Proc. 2nd USENIX HotCloud*.
- Zaharia, M., et al. (2012). Resilient distributed datasets: A fault-tolerant abstraction for in-memory cluster computing. *Proc. 9th USENIX NSDI*, 15–28.

Textbooks [SA], [TE], [TW]

- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 6 — Apache Spark). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 8 — In-Memory Processing). Prentice Hall.
- **[TW]** White, T. (2015). *Hadoop: The Definitive Guide*, 4th ed. (Ch. 19 — Spark). O'Reilly.

Case Study Source

- Databricks Engineering Blog. (2013, November). *Spark Officially Sets a New Record in Large-Scale Sorting*. databricks.com/blog/2013/11.

Closing — Week 6, Part 2

Key Takeaways

- Zaharia et al. (2012) showed that **lineage is a superior fault-tolerance mechanism** for iterative in-memory workloads — cheaper than replication, and it enables lazy evaluation
- The **narrow / wide dependency** distinction directly determines Spark's stage structure: minimising wide dependencies is the single most important Spark performance tuning heuristic
- Databricks' GraySort result (32× better efficiency per node) proved that **algorithm + hardware co-design** — sort-based shuffle on SSDs — matters as much as in-memory caching
- RDDs are now largely replaced by **DataFrames and Datasets** (since Spark 2.0), but every Spark DataFrame operation compiles down to an RDD plan at execution time

Quote of the Week

"RDDs can express many current cluster programming models — MapReduce, DryadLINQ, Pregel, HaLoop — as Spark libraries."

— Zaharia et al. (2012)

Part 1 Preview — Week 7

HiveQL, Partitioning, Bucketing & Presto

- HiveQL execution plan walkthrough
- Partitioning vs. bucketing trade-offs
- Presto/Trino federation model
- Cost-based optimiser and statistics